person. Because of this, DNA segment of third person will be larger in size followed by DNA segment of first person and then the second person. Thus, it is clear that satellite DNA bring about variation within the population. Variation in DNA banding pattern reveals differences among the individuals.

### Applications of DNA Fingerprinting

i. DNA fingerprinting technique is widely used in forensic applications like crime investigation such as identifying the culprit. It is also used for paternity testing in case of disputes.
ii. It also helps in the study of genetic diversity of population, evolution and speciation.

## 20.9 Genetically Modified Organisms (GMOs)

One of the most tremendous development of genetic engineering is the production of genetically modified (GM) plants and animals. **Genetic modification** refers to the alteration or manipulation of genes in the organisms using rDNA techniques in order to produce the desired characteristics. The DNA fragment inserted is called **transgene**. Plants or animals expressing a modified endogenous gene or a foregin gene are also known as **transgenic organisms**.

The transgenic plants are much stable, with improved nutritional quality, resistant to diseases and tolerant to various environment conditions. Similarly transgenic animals are used to produce proteins of medicinal importance at low cost and improve livestock quality.

Some examples of genetically modified plants and animals are given in the table below.

**Genetically Modified Plants**

| Objective | Gene inserted | Achievement |
|---|---|---|
| Improved nutritional quality in Rice | Beta carotene gene (In humans, Beta carotene is required for the synthesis of Vitamin A) | **Golden Rice** (Genetically modified rice can produce beta carotene, that can prevent Vitamin A deficiency) |
| Increased crop production | Bt gene from bacteria *Bacillus thuringiensis*. (Bt gene produces a protein that is toxic to insects) | **Insect resistant plants** (These plants can produce the toxin protein that kills the insects which attack them) |

**Genetically Modified Animals**

| Objective | Gene inserted | Achievement |
|---|---|---|
| Improved wool quality and production | Genes for synthesis of amino acid, cysteine | **Transgenic sheep** (gene expressed) |
| Increased growth in fishes | Salmon or Rainbow trout or Tilapia growth hormone gene | **Transgenic fish** (gene expressed) |



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